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Group Project Paper

Introduction

The original goal of this project was to design a device that aided the elderly in getting up and down stairs. As the project progressed, the focus changed to carrying items up and down the stairs. There were three main phases: the foundational support work, the feasibility prototype, and the functional prototype.

Preparation

To provide a foundation for our project, we sought to first empathize with our prospective clients. This involved conducting three interviews with elderly people. We discovered from these interviews that the main barriers to traversing stairs were the effects of previous injury, loss of muscle mass and eyesight due to age, the conditions of the stairs, and carrying objects. These concerns gave the team inspiration to work on a variety of designs. Some ideas were a walker intended for stair use, a miniature ski lift seat, and arm sleeves that gripped the handrail. However, the designs produced lacked focus and confidence or were products already in circulation. Therefore, we decided to narrow our focus to a specific challenge: handling objects while traveling on stairs. The two designs we focused on were a collapsible laundry chute

[[Figure 1](#)] and a box-shuttle design [[Figure 2](#)], the latter becoming our final design. The chute design was discarded because it was too specific and could not bring loads back up the stairs. On the other hand, the box-shuttle design showed promise because it was adaptable and could bring loads up and down the stairs. To solidify our decision, we created virtual prototypes in Solidworks and consulted with our customers. They also preferred the box-shuttle design.

To iron out the details of the project, we split the design into four parts, the shuttle, the cleat, the adjustor, and the pulley system. We separated into focus groups and took an hour to model each piece, coming back together to consult on what worked and what didn't. This allowed us to quickly prototype, test, and decide on different designs. After we completed the process, it gave us the design that would be used in the feasibility prototype.

Feasibility Prototype

For our feasibility prototype, we used a 2x4 beam as our track and designed a shuttle of wood that went around the beam. Half of the cleat was drilled into the shuttle, and the other half was connected to a platform. We tied a string to the side of the shuttle, which would be attached to a hand drill that acted as a motor at the top of the track. When activated, the hand drill would wind up the string and pull the shuttle up. To get the shuttle back down the track, we would spin the drill in the opposite direction which gave the line slack and allowed gravity to pull it down. Occasionally, the friction caused by the load would make it difficult for the shuttle to slide down. We used a hand drill as our motor because buying a real motor would put us well over our budget. The drill worked fairly well, despite the speed of its rotation being difficult to control. The cleat we used was also not very stable and couldn't hold much weight without slipping off. The track was held up with a beam on each side, one taller than the other to make the track at an

approximate 20-degree angle. We also made an adjuster panel, laser printed from MDF, that went between the shuttle and cleat which allowed us to adjust the angle of the platform on different angles of staircases [[Figure 4](#)]. This is a component we kept for the functional prototype, although we did make it larger and altered it slightly. The feasibility prototype helped us focus on what needed to be worked on the most, and what major problems we had. Firstly, we needed a track and shuttle with much less friction. Secondly, we also needed to redesign the cleat so it could support more weight. Lastly, both the upward and downward movements had to be powered to provide better control and boost efficacy. Using the materials available to us, which were mostly low-quality wood, the cleat itself was quite heavy, so we tried to minimize the weight of it for the functional prototype.

Functional Prototype

After identifying what we liked and what needed improvement from the feasibility prototype, we chose the final parts and what redesigns would need to be implemented. The first decision of many was what we wanted for our track design. Our initial, small-scale prototype, which would have utilized two parallel steel rods, was a good design in concept, but we decided that manufacturing such a part to the quality we desired was not within our capability, so we began to look for other preexisting track designs. Coming from our feasibility prototype, we knew that the mechanism would need to involve a sliding element, as it would provide the most torsional resistance. We decided to order an aluminum track from a website called 80/20. We purchased two 48-inch lengths. The manufacturer of the track had purchasable inserts, but we decided that we would design our own based on the engineering drawings of the track that they provided.

Once the tracks had arrived we could test the sliders we 3D printed shown in [\[Figure 3\]](#). The results told us which design worked best and based on that, we printed four of them. In order to attach the track to a frame that would hold it up at the angle of a staircase, we drilled two holes on each side with a precise vertical mill in the area of the track where the slider head would go. We drilled holes with the same separation in two sections of a 2x4 beam, so we could bolt the track to the beams. We ensured the distance between the tracks matched the sliders bolted to the slider panel.

Our previous French cleat design struggled to support a load that was concentrated towards the end of the platform and would slip out from the cleat hang from the moment. The functional cleat was remade as shown in figure 8 [\[Figure 8\]](#). We changed it significantly from the feasibility prototype. First, instead of an angled cleat, we made a slot and receptor that fit into each other [\[Figure 9\]](#). We also added two small triangular supports below that help support the platform. The actual platform we made was 12x20". If we were ever to make a full-scale, market-ready product, we would have preferred to make a platform with a T-shape so it would be lightweight and easier to use.

For our final pulley design, we identified we needed a way to power it while it went up and down. This is because the increased load would cause too much friction for gravity to pull the shuttle down. Powering the downward movement would also provide a more controlled descent. To do this, we attached a $\frac{1}{2}$ " diameter aluminum rod to the hand drill and moved it to the bottom of the assembly. Next, we wrapped two nylon cords in different directions around the rod. When the drill would spin counterclockwise, one cord would wind and the other would unwind, and it would reverse when it spun clockwise. One of the chords was 4 feet long and it was wrapped from the rod at the bottom to the shuttle. The other was 8 feet long and wrapped

from the rod at the bottom to the pulley at the top, and then back down to the other side of the shuttle. This gave the shuttle the ability to move up and down the rails in a controlled manner.

Conclusion

While we are incredibly proud of our result and aware it represents dozens of hours of work, there are things we would change before putting it on the market. For example, we would consider replacing the towers at both ends with more frequent steel supports driven into the surface of the tread. Instead of a drill, we would use a motor that better accommodates the design challenges. We would also make cosmetic changes. The rods used to coil the rope pulley and pin the track to the tower could be significantly shortened to stay out of the way and made thicker to increase torque. Some things we liked about our final product are the attachment of the track and shuttle, and the removability of the platform. Making the platform have variable components would accommodate customers' needs and widen the market. Having interviewed our clients one last time, their feedback concurred with our observations: The device works but it needs to be a little more user-friendly.

Overall, our project was a success. It can hold the weight of a typical load of laundry, the shuttle goes up and down a diagonal track, and the whole system isn't an impediment to regular stair use. It is a safe and effective device to aid in the navigation of stairs. A significant reason for our success is each team member's commitment to carry their weight and contribute meaningfully. It was a positive group work experience, and the team dynamic was collaborative and communicative. We learned a lot about using vertical mills, regular shop equipment, and Solidworks. We learned how to effectively communicate our ideas through sketches, interviews,

team conversations, and presentations. We are incredibly grateful for the experience this project and Mr. Pierson provided us and the skills we acquired along the way.

Appendix

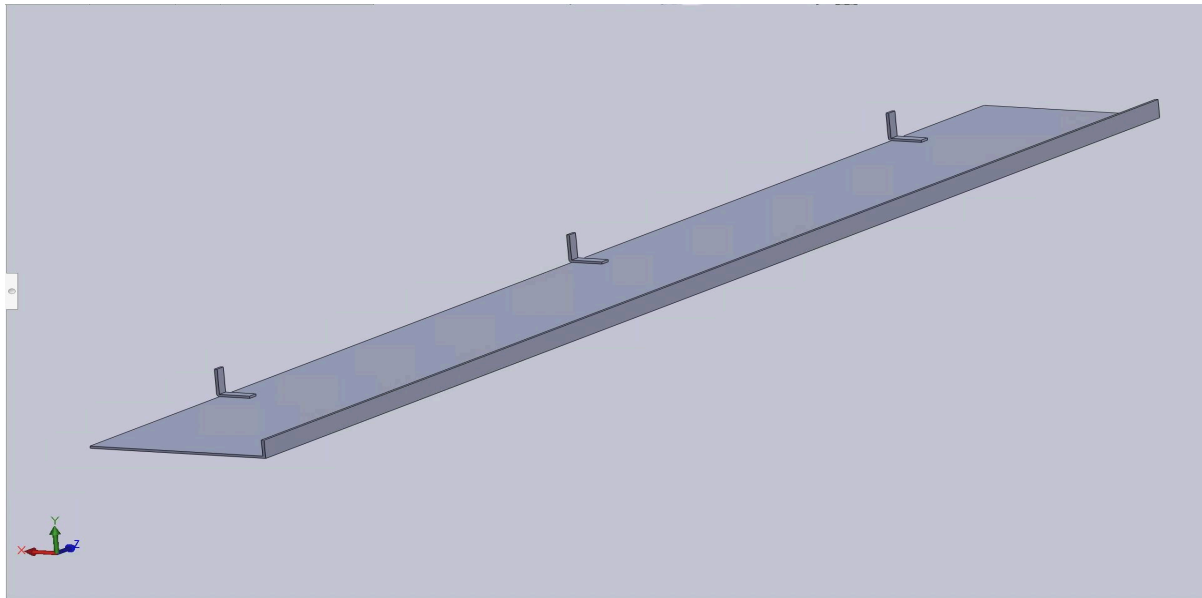


Figure 1: Laundry Chute

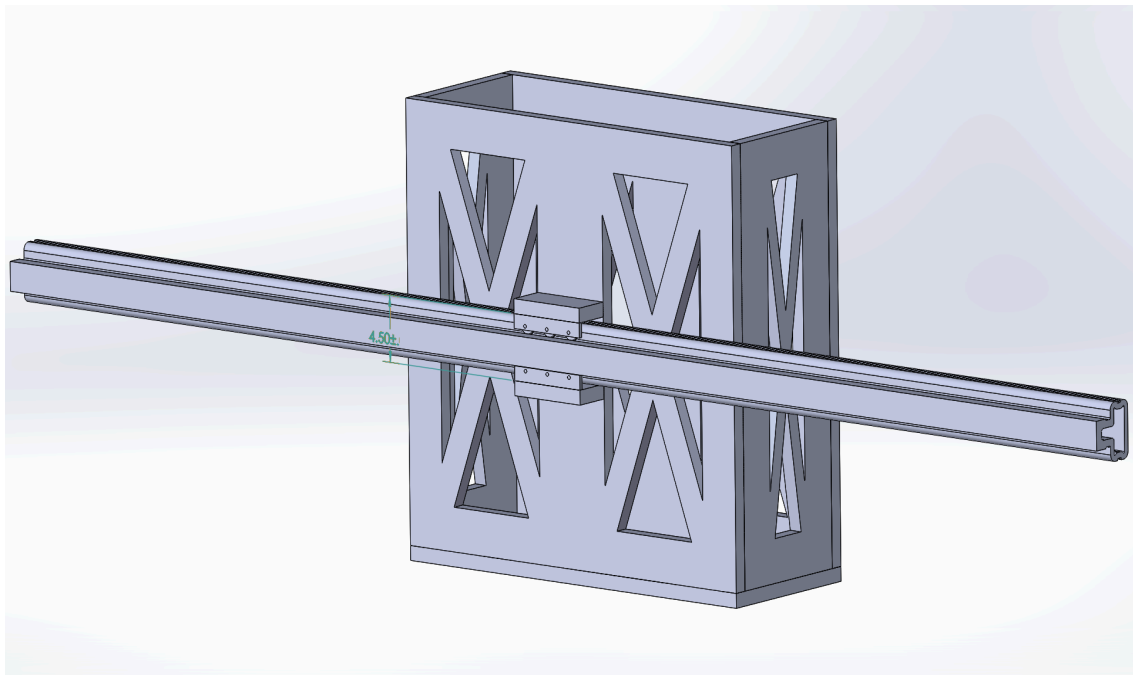


Figure 2: Box-Shuttle

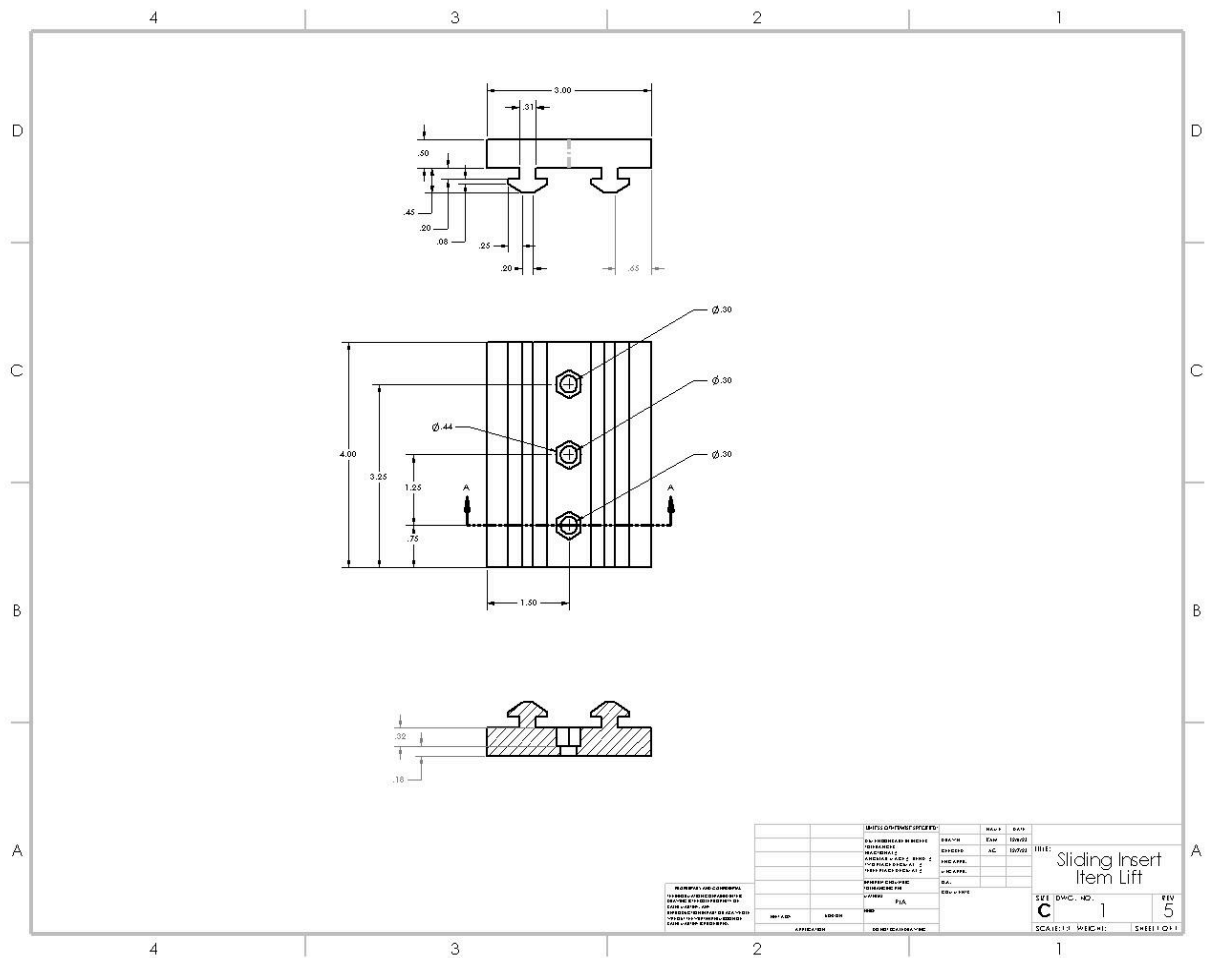


Figure 3: Slider of the shuttle

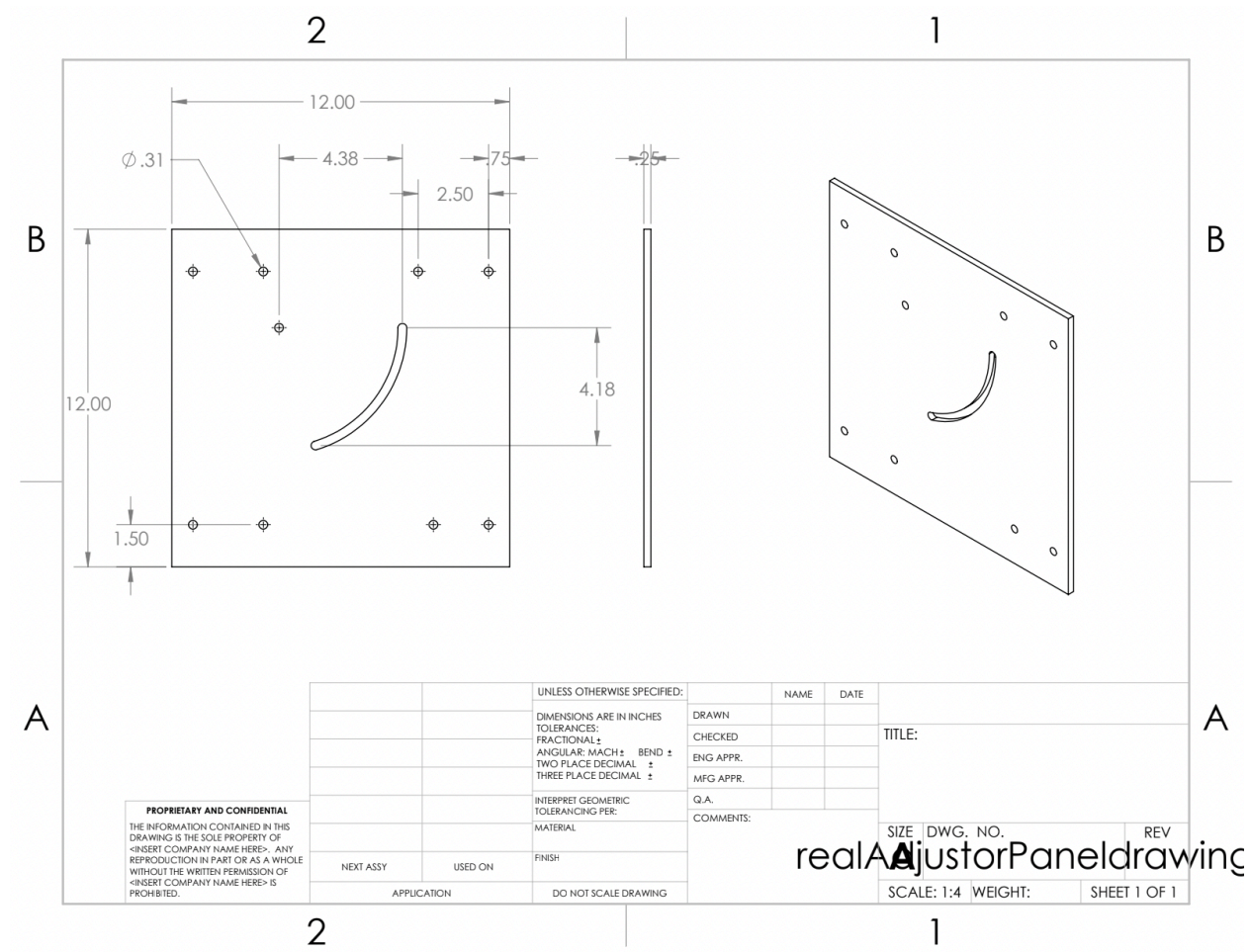


Figure 4: Adjustor Panel

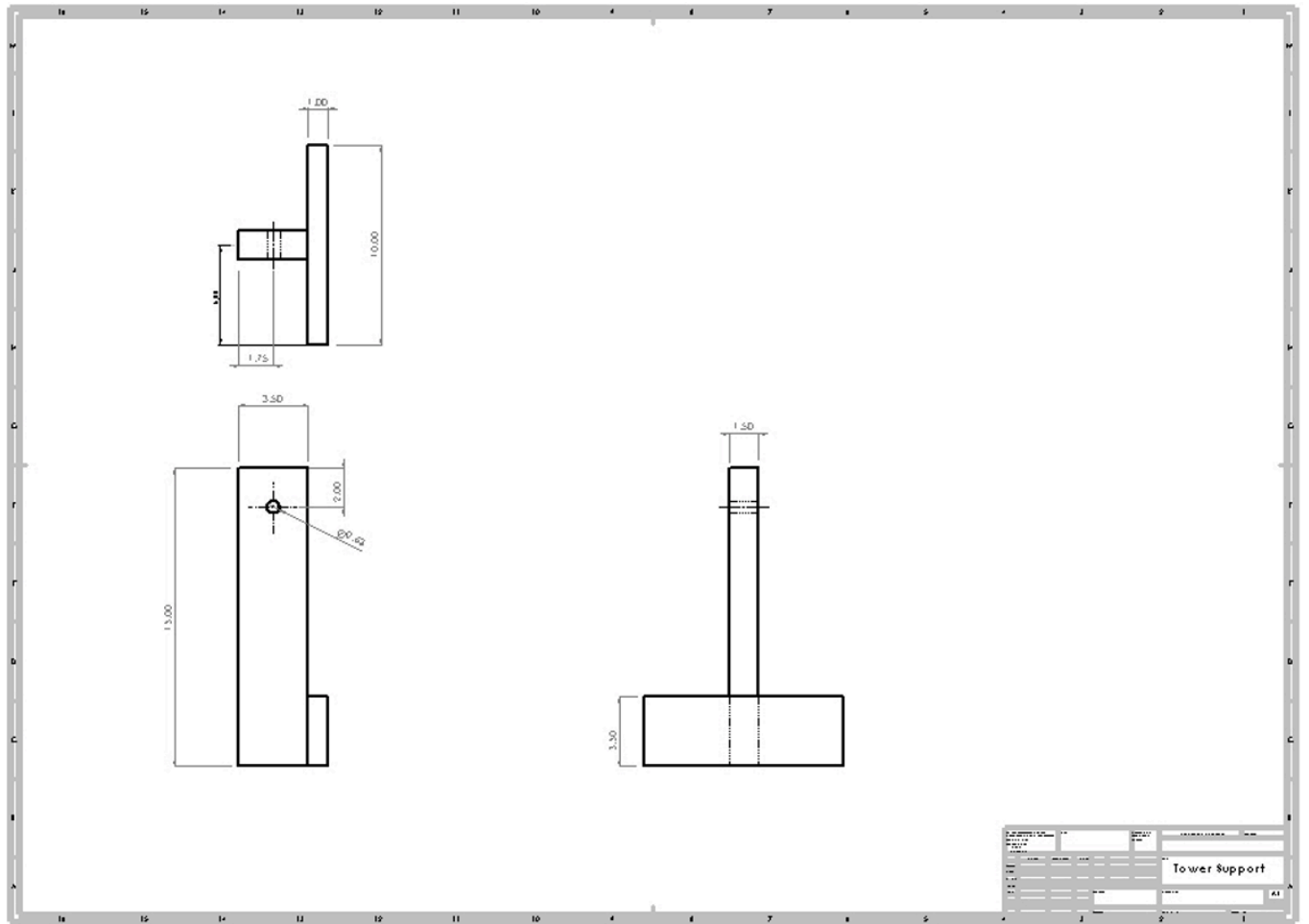


Figure 5: Tower Support

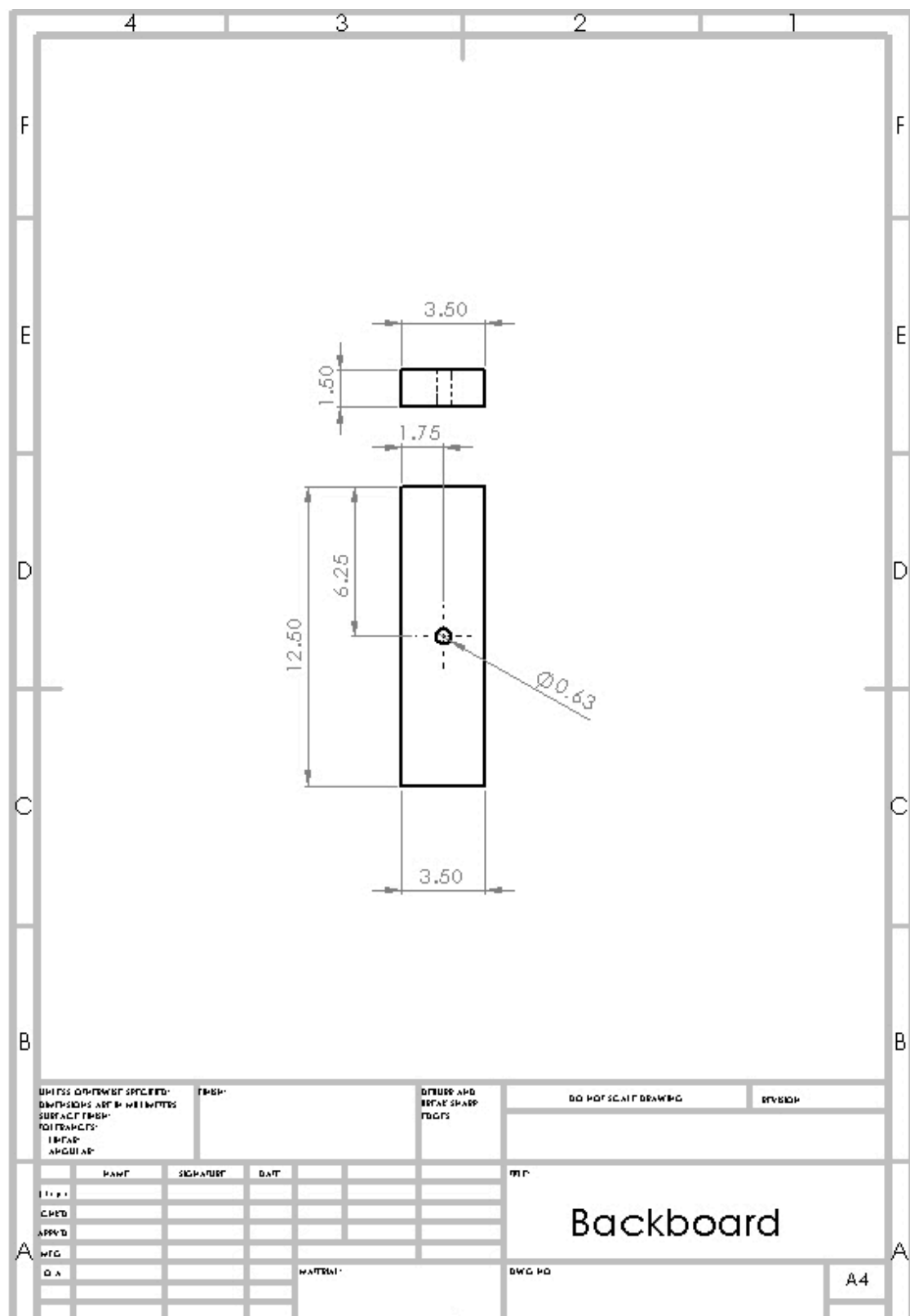


Figure 6: Back Board

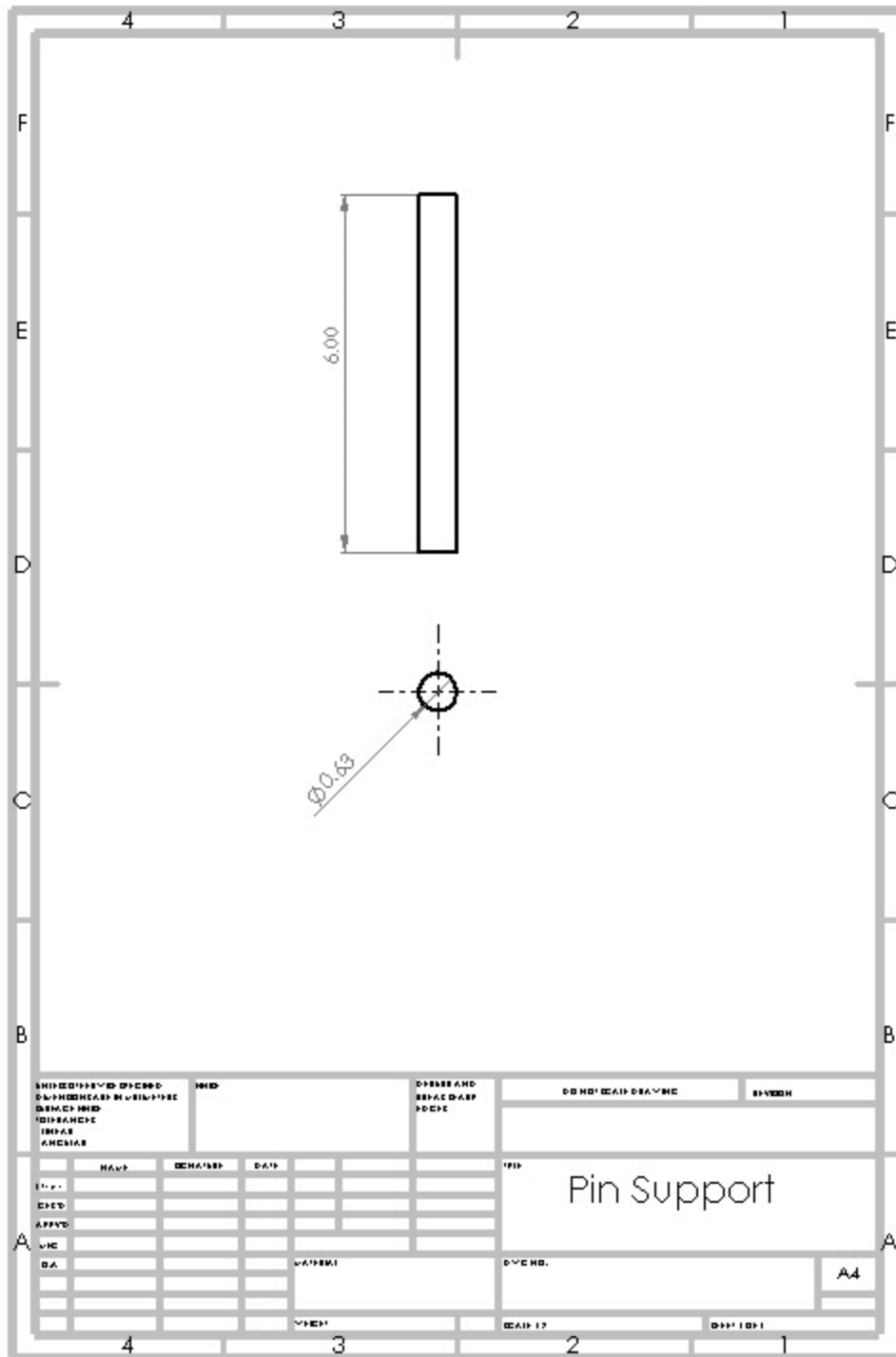


Figure 7: Pin Support

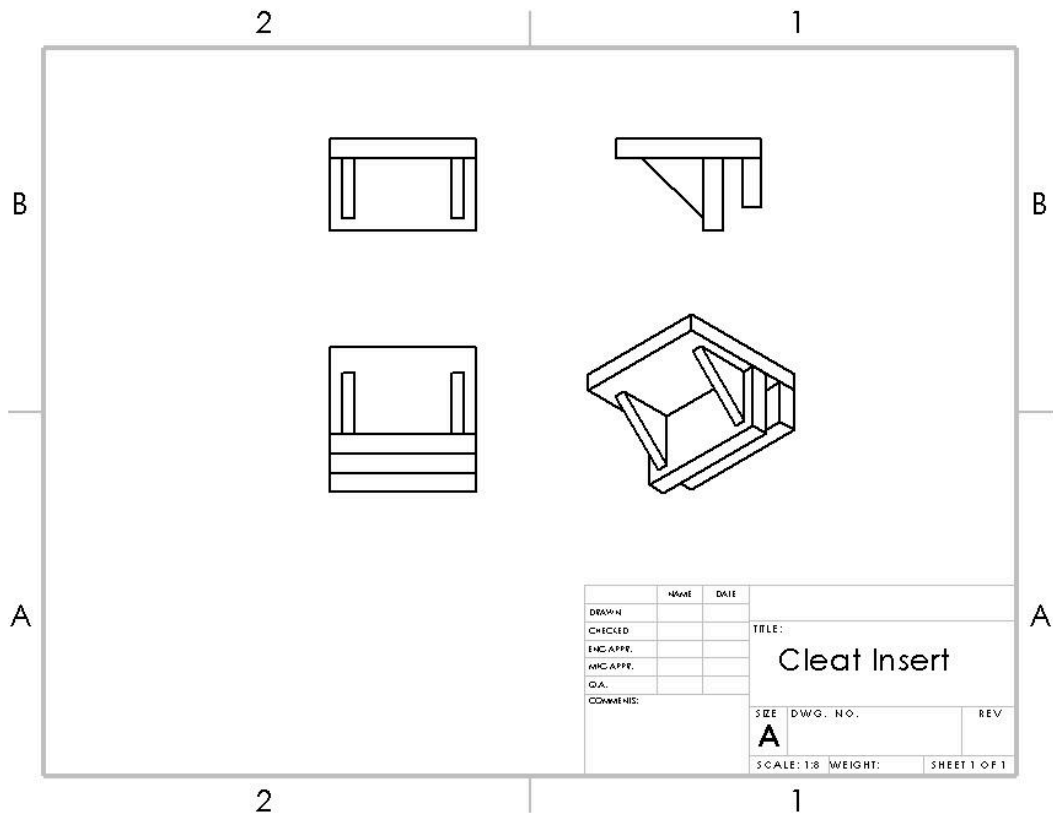


Figure 8: Cleat Assembly

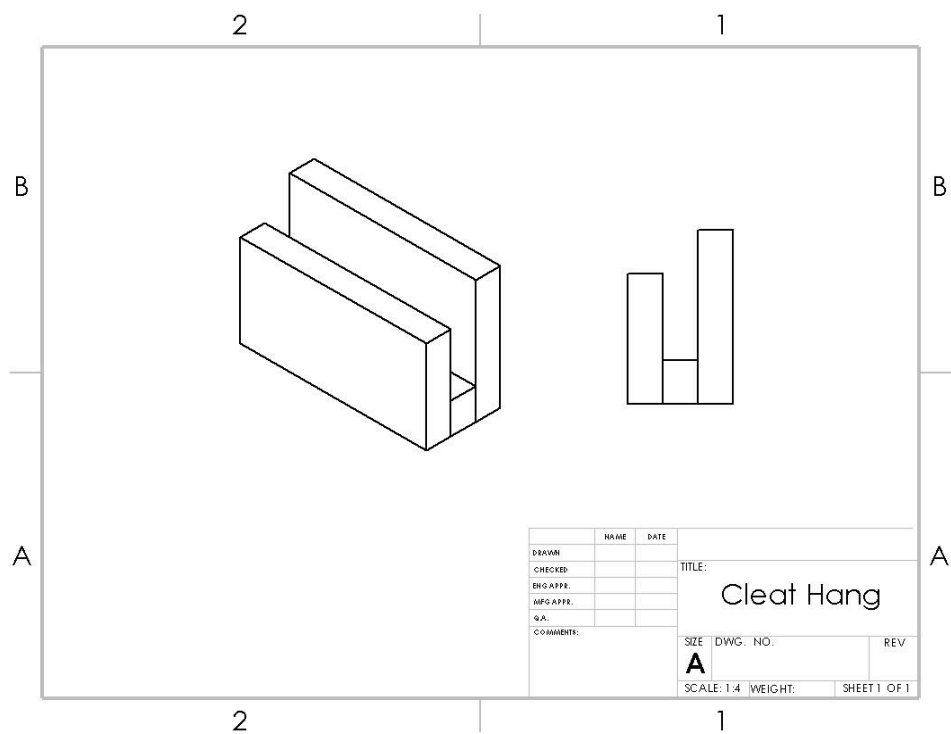


Figure 9: Receiving half of the cleat

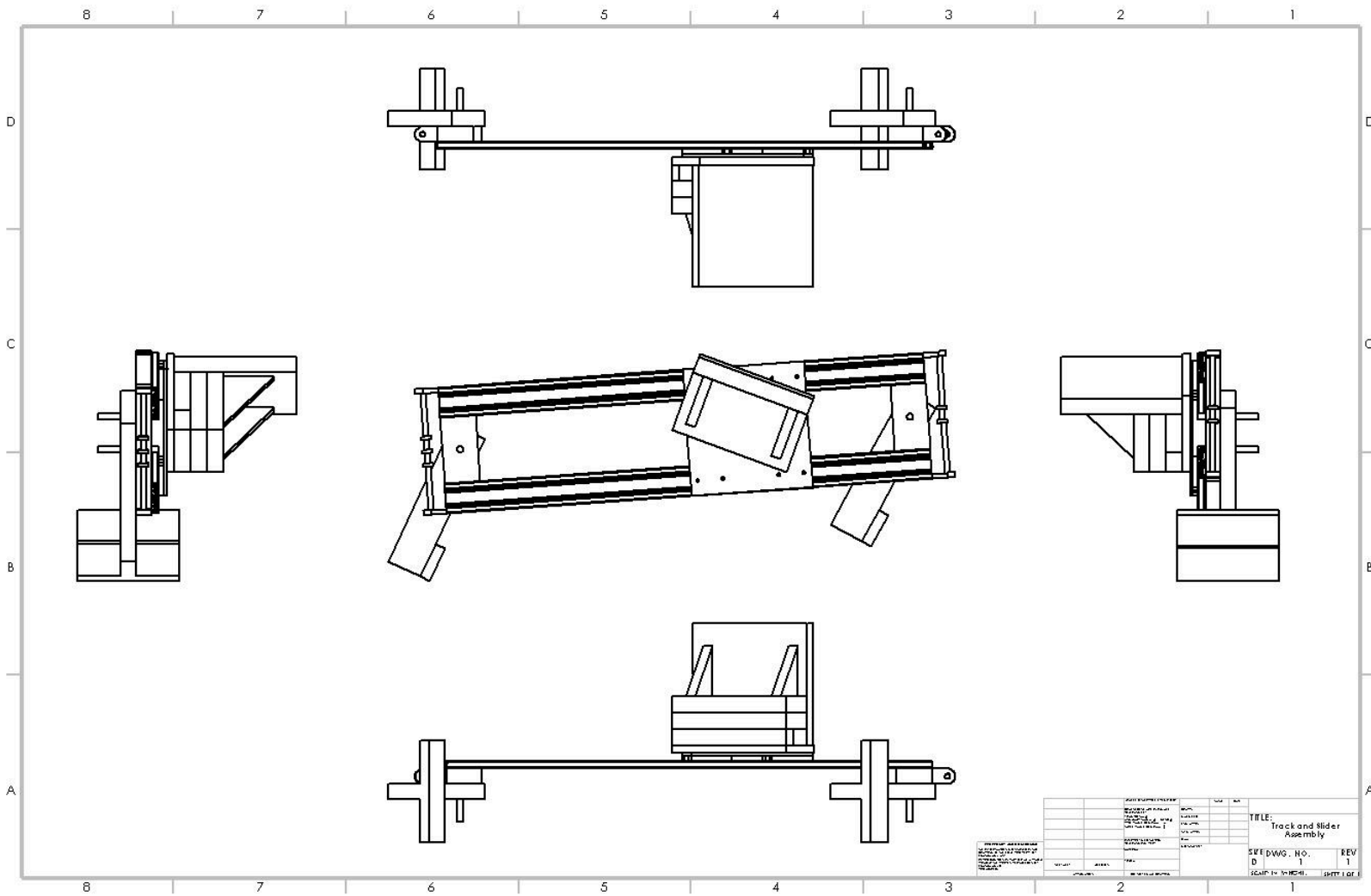


Figure 9: Full Assembly

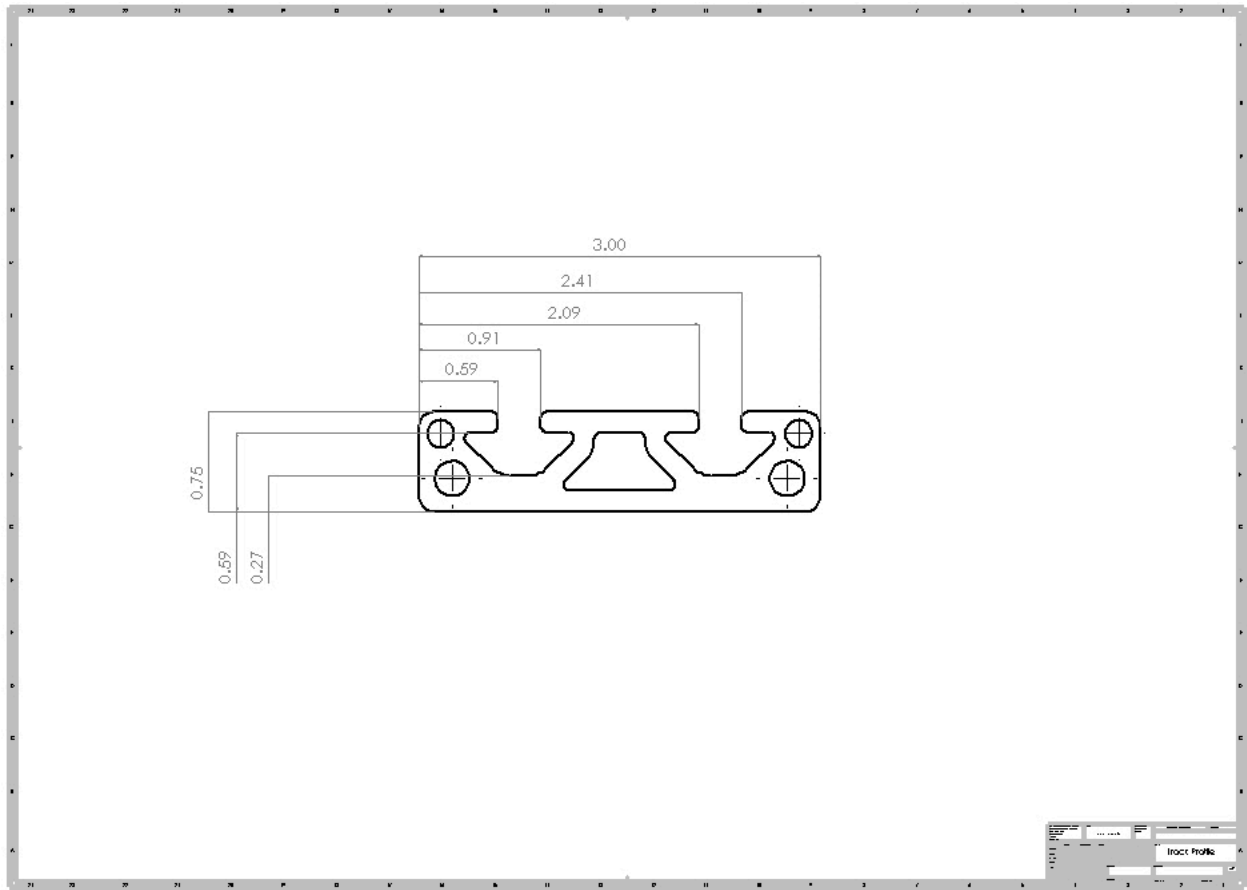


Figure 10: Track Profile

Part name	Dimensions	Material	Finish	Color	Quantity	Net Cost
Tower Post	15"	2x4" Wood	Treated	Tan	2	Free (storage room)
Tower base	10"	1x3.5" Wood	Treated	Tan	2	Free (storage)
Track	4'	Aluminum	—	Light gray	2	\$150 (includes shipping)
Back Board	12.5"	2x4" Wood	Treated	Tan	2	Free (storage)
Adjuster Panel	12x12"	MDF	N/A	Tan	2	Free (ENG Lab)
Sliders	4x3"	CPE	N/A	Black	4	Free (ENG Lab)
Cleat	12x20"	Wood	Treated	Tan	1	Free (ENG Lab)
Axle	12.5"	Steel	—	Grey	1	Free (ENG Lab)
Motor (Drill)	N/A	N/A	N/A	Green	1	Free (ENG Lab)
Cord	12'	Nylon	N/A	White	2	Free (ENG Lab)
Pulley	1/2"	Steel	—	Grey	1	Free (storage room)
Partially Thread Bolts	5/16 x 3 1/2"	Steel	Zinc	Grey	8	\$5 (approx.)
Misc screws, washers	As needed	Steel	Zinc	Grey	As needed	Free (ENG Lab)
Pin support	—	Steel	plain	Grey	2	Free (storage room)